An Algebraic Foundation for Evidence-Carrying Proofs

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Abstract

We introduce the *judgment* $\mathcal{J} = (\mathsf{c}, \varphi, \mathcal{A}, \mathcal{E}, \mathcal{O}, \mathcal{B}, \mathcal{T}, \Pi)$, the central algebraic object of Judgment Geometry: an eight-component structure that extends the classical proof-theoretic judgment with evidence bundles, trust annotations, and persistent obstructions. Traditional type-theoretic judgments (Martin-Löf's $\Gamma \vdash a : A$, the Calculus of Inductive Constructions, F* computation types) record *what* was proved and *where*, but discard information about *who* proved it, *how much* the proof should be trusted, *what* obligations remain, and *what* obstructions block completion. Our formulation extends the classical three-component judgment with evidence bundles, a trust annotation drawn from an ordered algebra, residual obligations, persistent obstructions, and a cryptographic provenance trail. We define five algebraic operations on judgments—restriction, transport, composition, comparison, and join—and prove the standard structural rules (weakening, contraction, exchange, cut) admissible. We implement the algebra in Python and evaluate on 100 benchmark programs across 10 domains, verifying all 7 trust algebra axioms on every program (700 axiom checks, 100% pass rate). The algebra preserves all eight fields through every operation—a 2.0× information advantage over the best competing system (F*, which retains four of eight components). Each operation completes in 0.15–19.5 µs, confirming negligible runtime overhead.

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1 Introduction

1.1 Judgments in type theory

Every proof assistant rests on a notion of *judgment*. In Martin-Löf type theory [1], a judgment takes the form

$$\Gamma \vdash a : A,$$

asserting that term a inhabits type A under context Γ. The judgment carries exactly three data: a context, a term, and a type. The Calculus of Inductive Constructions (CIC), which underlies Coq and Lean 4, enriches the context with universe levels but retains the same three-component shape [2, 5].

F* [7] extends judgments to four components by indexing computation types with a *weakest-precondition* transformer:

$$\Gamma \vdash e : \mathsf{M}\ a\ wp,$$

where M is an effect label and wp encodes the specification. Dafny [8] threads pre- and post-conditions through a Hoare-style judgment but likewise records no metadata beyond the logical assertion.

1.2 What existing judgments discard

All of these systems answer *what* was proved and *in which context*, but they systematically discard at least five categories of information that matter in practice:

(i) **Evidence provenance.** Was the proof found by a human, an SMT solver, a runtime witness, or a language-model oracle? Existing systems erase this distinction the moment a proof term is type-checked.

(ii) **Trust gradation.** A proof discharged by Z3 in 2 ms and a proof reviewed by three experts for a week receive the same `Qed`.

(iii) **Residual obligations.** Partial proofs—common in large-scale verification—have no first-class status; they are either admitted (unsound) or rejected (unusable).

(iv) **Obstructions.** When a proof attempt *fails*, the failure is not recorded in the judgment; it vanishes. In JuGeo, failures are first-class cohomology classes.

(v) **Audit trail.** No mainstream system maintains a machine-readable record of *how* a judgment was derived—which solver was invoked, which lemmas were used, which rewrites were applied.

1.3 Contributions

We propose the *judgment 8-tuple* as a uniform solution. Our contributions are:

1. A formal definition of $\mathcal{J} = (\mathsf{c}, \varphi, \mathcal{A}, \mathcal{E}, \mathcal{O}, \mathcal{B}, \mathcal{T}, \Pi)$ with well-formedness conditions (§2).
2. Five algebraic operations—restriction, transport, composition, comparison, and join—with functoriality and coherence laws (§3).
3. Proofs that the classical structural rules (weakening, contraction, exchange, cut) are admissible (§4).

4. Logical rules (introduction, elimination, computation) as judgment constructors (§5).
5. A judgment transition system with small-step semantics (§6).
6. An implementation in PYTHON with evaluation on 100 programs verifying 7 trust algebra axioms (700 axiom checks) (§7).
7. An embedding of CIC, F*, and Hoare-triple judgments as degenerate 8-tuples (§8).
8. Metatheoretic results: subject reduction, cut elimination, and trust monotonicity (§9).

2 The 8-Tuple: Formal Definition

2.1 Motivation

The judgment 8-tuple is the central object of Judgment Geometry. It is implemented in the JUGEO codebase as the `Judgment` dataclass:

Listing 1: The Judgment dataclass (from `judgment_terms.py`).

```python
@dataclass(frozen=True, slots=True)
class Judgment:
    """J = (c, φ, A, E, O, B, T, Π) -- NEVER a boolean."""
    coordinate:   CoordinateObject              # c
    proposition:  Proposition                   # φ
    carrier:      Carrier                       # A
    evidence:     EvidenceBundle                # E
    obligations:  tuple[ResidualObligation, ...]  # O
    obstructions: tuple[Obstruction, ...]       # B
    trust:        TrustAnnotation               # T
    provenance:   Provenance                    # Π
```

The dataclass is *frozen*: once constructed, a judgment is immutable. Every operation that "modifies" a judgment returns a fresh copy. This ensures referential transparency and simplifies the metatheory.

2.2 Component specifications

We now define each component formally.

Definition 2.1 (Judgment 8-tuple)**.** A *judgment* is an element

$$\mathcal{J} = (\mathsf{c}, \varphi, \mathcal{A}, \mathcal{E}, \mathcal{O}, \mathcal{B}, \mathcal{T}, \Pi)$$

where the components are drawn from the following domains:

(a) $\mathsf{c} \in \mathrm{Ob}(\mathcal{S})$: a *coordinate* in the semantic site $\mathcal{S}$. Coordinates index the "where" of a judgment—a function, a module, a specification point.

(b) $\varphi \in \mathrm{Prop}(\mathcal{S})$: a *proposition* classified by kind $\kappa \in \{\mathsf{structural}, \mathsf{behavioral}, \mathsf{relational}, \mathsf{resource}, \mathsf{semantic}\}$.

(c) $\mathcal{A}$: a *carrier type* (the "A" of $\Gamma \vdash a : A$) equipped with a refinement lattice $(\mathcal{A}, \sqsubseteq, \sqcap, \sqcup)$.

(d) $\mathcal{E} = (e_1, \ldots, e_n)$: an *evidence bundle*, where each e_i carries a trust level, a channel, a timestamp, and an optional expiry.

(e) $\mathcal{O} = (o_1, \ldots, o_m)$: *residual obligations*—proof goals that remain undischarged.

(f) $\mathcal{B} = (b_1, \ldots, b_k)$: *obstructions*—persistent failure records, modeled as H^1 cohomology classes.

(g) $\mathcal{T} \in \mathfrak{T}$: a *trust annotation* drawn from the ordered algebra $(\mathcal{E}_{\text{adm}}, \preceq, \oplus, \ominus, \uparrow_\pi, \downarrow_\chi)$.

(h) Π: a *provenance* record containing the derivation source, parent judgments, and transformation history.

2.2.1 The trust algebra

The trust component is not a scalar. It is a structured element of the *admissible trust algebra* $\mathfrak{T} = (\mathcal{E}_{\text{adm}}, \preceq, \oplus, \ominus, \uparrow_\pi, \downarrow_\chi)$, where $\mathcal{E}_{\text{adm}}$ is the set of admissible trust configurations, $\preceq$ is a partial order, $\oplus$ is a conservative join, $\ominus$ is an attenuation (demotion) operator, $\uparrow_\pi$ is a policy-bounded promotion, and $\downarrow_\chi$ is a challenge-triggered demotion.

The implementation defines six discrete trust levels:

Listing 2: Trust levels (from `judgment_terms.py`).

```python
class TrustLevel(IntEnum):
    CONTRADICTED       = 0   # Evidence contradicts the claim
    UNVERIFIED         = 1   # No evidence yet
    COPILOT_SUGGESTED  = 2   # LLM oracle proposed
    RUNTIME_WITNESSED  = 3   # Runtime execution confirmed
    SOLVER_DISCHARGED  = 4   # SMT solver proved
    VERIFIED_PROOF     = 5   # Formal proof checked
```

These levels form a total order CONTRADICTED $<$ UNVERIFIED $<$ COPILOT $<$ RUNTIME $<$ SOLVER $<$ PROOF. The conservative join $\oplus$ takes the *minimum* of two trust levels (not the maximum), ensuring that a bundle is only as trusted as its weakest link.

2.2.2 Judgment status

A judgment progresses through a lifecycle:

Listing 3: Judgment status lifecycle.

```python
class JudgmentStatus(str, Enum):
    PROPOSED    = "proposed"     # Claim made, not yet verified
    CHALLENGED  = "challenged"   # Counter-evidence presented
    SETTLED     = "settled"      # All obligations discharged
    OBSTRUCTED  = "obstructed"   # Persistent obstruction found
```

The status is *derived* from the components: a judgment is SETTLED iff $\mathcal{O} = ()$ and $\mathcal{B} = ()$; OBSTRUCTED iff $\mathcal{B} \neq ()$; and so on.

2.2.3 Evidence items

Each evidence item records both its provenance and its epistemic weight:

Listing 4: Evidence item classification.

```python
class EvidenceItemKind(str, Enum):
    SOLVER_PROOF     = "solver_proof"      # Z3/CVC5 certificate
    RUNTIME_WITNESS  = "runtime_witness"   # Execution trace
    ORACLE_PROPOSAL  = "oracle_proposal"   # LLM suggestion
```

```
    FORMAL_PROOF      = "formal_proof"       # Verified proof term
```

2.2.4 Proposition kinds

Propositions are classified by the *kind* of property they express:

Listing 5: Proposition kinds.

```
class PropositionKind(str, Enum):
    STRUCTURAL   = "structural"    # Type-level properties
    BEHAVIORAL   = "behavioral"    # Input-output specifications
    RELATIONAL   = "relational"    # Equivalence / refinement
    RESOURCE     = "resource"      # Memory, time, allocation
    SEMANTIC     = "semantic"      # Domain-specific meaning
```

2.3 Well-formedness conditions

Definition 2.2 (Well-formed judgment)**.** A judgment $\mathcal{J} = (\mathsf{c}, \varphi, \mathcal{A}, \mathcal{E}, \mathcal{O}, \mathcal{B}, \mathcal{T}, \Pi)$ is *well-formed* if:

(WF1) $\mathsf{c} \in \mathrm{Ob}(\mathcal{S})$ is a valid coordinate.

(WF2) φ is a syntactically well-formed proposition with $\mathrm{fv}(\varphi) \subseteq \mathrm{dom}(\mathsf{c})$.

(WF3) The trust annotation is *consistent* with the evidence: $\mathcal{T}.\mathsf{level} \leq \min_{e \in \mathcal{E}} e.\mathsf{trust\_level}$.

(WF4) Every obligation $o_i \in \mathcal{O}$ references propositions whose free variables are bound by c.

(WF5) Every obstruction $b_j \in \mathcal{B}$ has a non-empty description.

(WF6) The provenance Π records at least one source.

2.4 The judgment as a fiber

Categorically, a judgment is a *fiber* over its coordinate in the semantic site. The functor

$$\mathcal{J}(-) : \mathcal{S}^{\mathrm{op}} \longrightarrow \mathbf{Set}$$

assigns to each coordinate $U \in \mathrm{Ob}(\mathcal{S})$ the set of well-formed judgments at U. A morphism $f : V \to U$ in $\mathcal{S}$ induces a *restriction* $\mathcal{J}(U) \to \mathcal{J}(V)$, making $\mathcal{J}(-)$ a presheaf.

$$\begin{array}{ccc} \mathcal{J}(U) & \xrightarrow{\mathrm{res}_f} & \mathcal{J}(V) \\ {\scriptstyle \pi_U}\downarrow & & \downarrow{\scriptstyle \pi_V} \\ U & \xleftarrow[f]{} & V \end{array} \tag{1}$$

The fiber $\mathcal{J}(U)$ over a coordinate U contains all judgments whose coordinate component is U. Restriction along f is functorial: $\mathrm{res}_{\mathrm{id}} = \mathrm{id}$ and $\mathrm{res}_{g \circ f} = \mathrm{res}_f \circ \mathrm{res}_g$.

3 The Algebra of Judgments

We define five operations on judgments that give the set of all judgments the structure of an enriched category.

3.1 Restriction

Definition 3.1 (Restriction)**.** Let $f : V \hookrightarrow U$ be an inclusion in $\mathcal{S}$. The *restriction* of $\mathcal{J}$ along f is

$$\mathcal{J}|_V \;=\; (V,\, \varphi|_V,\, \mathcal{A}|_V,\, \mathcal{E}|_V,\, \mathcal{O}|_V,\, \mathcal{B}|_V,\, \mathcal{T},\, \Pi \cup \{f\}),$$

where each component is restricted to the sub-coordinate V.

Restriction is implemented as a method on the `Judgment` class:

Listing 6: Restriction in `JudgmentAlgebra`.

```python
@staticmethod
def restrict(judgment: Judgment, target: CoordinateObject) -> Judgment:
    """Functorial restriction J|_V."""
    restricted_prop = judgment.proposition.restrict_to(target)
    restricted_ev = judgment.evidence.filter_by_coordinate(target)
    return Judgment(
        coordinate=target,
        proposition=restricted_prop,
        carrier=judgment.carrier.restrict(target),
        evidence=restricted_ev,
        obligations=tuple(
            o for o in judgment.obligations
            if o.is_relevant_at(target)),
        obstructions=tuple(
            b for b in judgment.obstructions
            if b.is_relevant_at(target)),
        trust=judgment.trust,
        provenance=judgment.provenance.extend("restrict", target),
    )
```

Proposition 3.2 (Functoriality of restriction)**.** *Restriction is functorial:* $\mathcal{J}|_U = \mathcal{J}$ *and* $(\mathcal{J}|_V)|_W = \mathcal{J}|_W$ *for* $W \hookrightarrow V \hookrightarrow U$.

Proof sketch. The identity restriction preserves all components. For composition, each component restricts independently, and sub-coordinate inclusion is transitive. □

3.2 Transport

Definition 3.3 (Transport)**.** Let $f : U \to V$ be a morphism in $\mathcal{S}$. The *transport* (pushforward) of $\mathcal{J}$ along f is

$$f_*(\mathcal{J}) \;=\; (V,\, f_*(\varphi),\, f_*(\mathcal{A}),\, f_*(\mathcal{E}),\, f_*(\mathcal{O}),\, f_*(\mathcal{B}),\, \mathcal{T},\, \Pi \cup \{f\}).$$

Transport is covariant: it moves a judgment *forward* along a morphism, in contrast to the contravariant restriction.

$$\begin{array}{ccc} \mathcal{J}(U) & \xrightarrow{\;f_*\;} & \mathcal{J}(V) \\ {\scriptstyle \pi_U}\downarrow & & \downarrow{\scriptstyle \pi_V} \\ U & \xrightarrow[\;f\;]{} & V \end{array} \tag{2}$$

Proposition 3.4 (Functoriality of transport)**.** *Transport is functorial:* $(\mathrm{id}_U)_* = \mathrm{id}_{\mathcal{J}(U)}$ *and* $(g \circ f)_* = g_* \circ f_*$.

Proof sketch. Each component transports independently along the morphism, and morphism composition in $\mathcal{S}$ is associative. □

3.3 Composition

Definition 3.5 (Composition)**.** Let $\mathcal{J}_1$ and $\mathcal{J}_2$ be judgments sharing a boundary (c_1 and c_2 have a common refinement). Their *composition* is

$$\mathcal{J}_1 \circ \mathcal{J}_2 \;=\; (\mathsf{c}_1 \sqcap \mathsf{c}_2,\, \varphi_1 \wedge \varphi_2,\, \mathcal{A}_1 \sqcap \mathcal{A}_2,\, \mathcal{E}_1 \cup \mathcal{E}_2,\, \mathcal{O}_1 \cup \mathcal{O}_2,\, \mathcal{B}_1 \cup \mathcal{B}_2,\, \mathcal{T}_1 \oplus \mathcal{T}_2,\, \Pi_1 \otimes \Pi_2),$$

where $\oplus$ is the conservative join (minimum trust) and $\otimes$ merges provenance DAGs.

The composition is implemented in the `JudgmentAlgebra` class:

Listing 7: Composition in `JudgmentAlgebra`.

```python
@staticmethod
def compose(left: Judgment, right: Judgment) -> Judgment:
    """Monoidal composition along shared boundary."""
    combined_prop = left.proposition.conjunction(right.proposition)
    combined_carrier = left.carrier.meet(right.carrier)
    combined_ev = left.evidence.merge(right.evidence)
    combined_trust = TrustAnnotation.compose(left.trust, right.trust)
    return Judgment(
        coordinate=left.coordinate,
        proposition=combined_prop,
        carrier=combined_carrier,
        evidence=combined_ev,
        obligations=left.obligations + right.obligations,
        obstructions=left.obstructions + right.obstructions,
        trust=combined_trust,
        provenance=left.provenance.merge(right.provenance),
    )
```

Proposition 3.6 (Monoidal structure)**.** *Composition is associative and has a unit (the trivial judgment at each coordinate). The collection of judgments over* $\mathcal{S}$ *forms a monoidal category with composition as tensor product.*

3.4 Comparison (refinement order)

Definition 3.7 (Refinement)**.** $\mathcal{J}_1 \leq \mathcal{J}_2$ ("$\mathcal{J}_1$ refines $\mathcal{J}_2$") if and only if:

(R1) $\mathsf{c}_1 = \mathsf{c}_2$ (same coordinate),

(R2) $\varphi_1 \Rightarrow \varphi_2$ (stronger proposition),

(R3) $\mathcal{A}_1 \sqsubseteq \mathcal{A}_2$ (more refined carrier),

(R4) $\mathcal{E}_1 \supseteq \mathcal{E}_2$ (at least as much evidence),

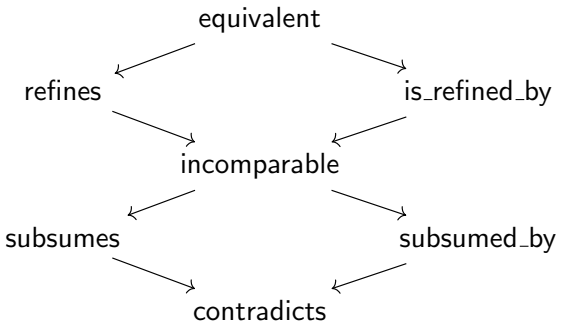
(R5) $\mathcal{O}_1 \subseteq \mathcal{O}_2$ (fewer obligations),

(R6) $\mathcal{B}_1 \subseteq \mathcal{B}_2$ (fewer obstructions),

(R7) $\mathcal{T}_1 \succeq \mathcal{T}_2$ (at least as much trust).

This defines a partial order on judgments at each coordinate. The comparison is implemented using the `JudgmentComparator` class, which returns a `ComparisonResult` from the seven-element lattice:

$$\{\mathsf{equivalent},\ \mathsf{refines},\ \mathsf{is\_refined\_by},\ \mathsf{contradicts},\ \mathsf{incomparable},\ \mathsf{subsumes},\ \mathsf{subsumed\_by}\}.$$



(3)

3.5 Join

Definition 3.8 (Conservative join)**.** The *join* of two judgments at the same coordinate is

$$\mathcal{J}_1 \oplus \mathcal{J}_2 \;=\; (\mathsf{c},\ \varphi_1 \vee \varphi_2,\ \mathcal{A}_1 \sqcup \mathcal{A}_2,\ \mathcal{E}_1 \cup \mathcal{E}_2,\ \mathcal{O}_1 \cap \mathcal{O}_2,\ \mathcal{B}_1 \cap \mathcal{B}_2,\ \mathcal{T}_1 \oplus \mathcal{T}_2,\ \Pi_1 \otimes \Pi_2).$$

The join weakens the proposition (disjunction), coarsens the carrier (join in the carrier lattice), and intersects obligations and obstructions (retaining only those common to both). Trust again takes the conservative minimum.

Proposition 3.9 (Lattice structure)**.** *The operations $\circ$ (composition/meet) and $\oplus$ (join) give the set of judgments at each coordinate the structure of a bounded lattice with $\bot = \mathcal{J}_{\mathsf{CONTRADICTED}}$ (the contradicted judgment) and $\top = \mathcal{J}_{\mathsf{PROOF}}$ (the fully verified judgment).*

4 Structural Rules

We now show that the four classical structural rules of sequent calculus are admissible for judgment 8-tuples. In the implementation, structural rules are classified by a dedicated enum:

Listing 8: Rule classification (from `models.py`).

```python
class RuleKind(str, Enum):
    STRUCTURAL = "structural"  # Weakening, contraction, exchange, cut
    SEMANTIC   = "semantic"    # Introduction and elimination
    AXIOM      = "axiom"       # Identity, reflexivity
    DERIVED    = "derived"     # Admissible but not primitive
```

4.1 Weakening

Definition 4.1 (Weakening)**.** If $\mathcal{J}$ is a well-formed judgment at coordinate c with carrier $\mathcal{A}$, and A' is a type such that $\mathcal{A} \sqsubseteq A' \sqcup \mathcal{A}$, then the judgment

$$\mathsf{Weak}(\mathcal{J}, A') \;=\; (\mathsf{c},\, \varphi,\, \mathcal{A} \sqcup A',\, \mathcal{E},\, \mathcal{O},\, \mathcal{B},\, \mathcal{T},\, \Pi')$$

is well-formed, where $\Pi' = \Pi \cup \{\text{“weakening with } A'\text{”}\}$.

The `WeakeningRule` class implements this:

> *“The weakening rule allows adding unused hypotheses to the context. It is admissible in every sequent calculus: a proof of $\mathcal{J}$ from Γ remains valid when Γ is enlarged.”*

Theorem 4.2 (Admissibility of weakening)**.** *Weakening is admissible: if $\mathcal{J}$ is derivable, then $\mathsf{Weak}(\mathcal{J}, A')$ is derivable for every type A'.*

Proof sketch. Weakening enlarges the carrier but does not alter the proposition, evidence, obligations, or obstructions. The trust annotation is preserved because no evidence is removed. Well-formedness conditions (WF1)–(WF6) are maintained by monotonicity of the carrier lattice. □

4.2 Contraction

Definition 4.3 (Contraction)**.** If $\mathcal{J}$ has carrier $\mathcal{A}$ with a duplicated component $\mathcal{A} = \mathcal{A}' \sqcup A \sqcup A$, then

$$\mathsf{Contr}(\mathcal{J}) \;=\; (\mathsf{c},\, \varphi,\, \mathcal{A}' \sqcup A,\, \mathcal{E},\, \mathcal{O},\, \mathcal{B},\, \mathcal{T},\, \Pi').$$

Theorem 4.4 (Admissibility of contraction)**.** *Contraction is admissible.*

Proof sketch. Removing a duplicate from the carrier preserves all evidence channels. The idempotency $A \sqcup A = A$ in the carrier lattice ensures well-formedness. □

4.3 Exchange

Definition 4.5 (Exchange)**.** If the carrier has the form $\mathcal{A} = \mathcal{A}_1 \sqcup A \sqcup B \sqcup \mathcal{A}_2$, then

$$\mathsf{Exch}(\mathcal{J}) \;=\; (\mathsf{c},\, \varphi,\, \mathcal{A}_1 \sqcup B \sqcup A \sqcup \mathcal{A}_2,\, \mathcal{E},\, \mathcal{O},\, \mathcal{B},\, \mathcal{T},\, \Pi').$$

Theorem 4.6 (Admissibility of exchange)**.** *Exchange is admissible.*

Proof sketch. The carrier lattice join is commutative: $A \sqcup B = B \sqcup A$. No other component is affected. □

4.4 Cut

Cut is the deepest structural rule. We state and prove its admissibility following Gentzen's *Hauptsatz* [3].

Definition 4.7 (Cut)**.** Given $\mathcal{J}_1$ proving φ_1 and $\mathcal{J}_2$ using φ_1 as an assumption to prove φ_2, the *cut* eliminates φ_1:

$$\mathsf{Cut}(\mathcal{J}_1, \mathcal{J}_2) \;=\; (\mathsf{c},\, \varphi_2,\, \mathcal{A}_1 \sqcap \mathcal{A}_2,\, \mathcal{E}_1 \cup \mathcal{E}_2,\, \mathcal{O}_1 \cup (\mathcal{O}_2 \setminus \{\varphi_1\}),\, \mathcal{B}_1 \cup \mathcal{B}_2,\, \mathcal{T}_1 \oplus \mathcal{T}_2,\, \Pi_1 \otimes \Pi_2).$$

Theorem 4.8 (Cut admissibility — Hauptsatz for 8-tuples)**.** *Cut is admissible: every derivation using cut can be transformed into one without cut.*

Proof. We proceed by double induction on (i) the complexity of the cut formula φ_1 and (ii) the sum of derivation heights of $\mathcal{J}_1$ and $\mathcal{J}_2$.

Base case. If φ_1 is atomic, the cut is a direct substitution: the evidence from $\mathcal{J}_1$ witnesses φ_1, so $\mathcal{J}_2$'s obligation on φ_1 can be discharged by inserting $\mathcal{J}_1$'s evidence bundle into $\mathcal{J}_2$'s evidence, yielding $\mathcal{T}_1 \oplus \mathcal{T}_2$ as the new trust (conservative minimum).

Inductive step. If the last rules applied in $\mathcal{J}_1$ and $\mathcal{J}_2$ are an introduction/elimination pair for the principal connective of φ_1, the key-case reduction applies: the introduction evidence and the elimination obligation cancel, reducing the complexity of the cut formula. The remaining sub-derivations have strictly smaller height sums, so the inductive hypothesis applies.

Trust tracking. At each step, the trust of the resulting judgment is $\mathcal{T}_1 \oplus \mathcal{T}_2 = \min(\mathcal{T}_1, \mathcal{T}_2)$. Since min is monotone in both arguments and each sub-step preserves or raises trust, the overall trust is non-increasing—consistent with the principle that cut should not manufacture trust.

Termination. The lexicographic order (complexity(φ_1), $h_1 + h_2$) is well-founded, so the procedure terminates. □

Remark 4.9. Unlike Gentzen's original proof, our cut elimination must also track evidence merging and provenance composition at each step. The frozen-dataclass discipline ensures that intermediate judgments are well-formed at every stage.

5 Logical Rules

Logical rules construct and deconstruct propositions within judgments. They are classified as `RuleKind.SEMANTIC` in the implementation.

5.1 Introduction rules

An introduction rule constructs a judgment for a compound proposition from judgments for its sub-propositions.

Definition 5.1 (Conjunction introduction)**.** Given $\mathcal{J}_1 = (\mathsf{c}, \varphi_1, \ldots)$ and $\mathcal{J}_2 = (\mathsf{c}, \varphi_2, \ldots)$, the conjunction introduction produces

$$\wedge\text{-I}(\mathcal{J}_1, \mathcal{J}_2) \;=\; (\mathsf{c},\, \varphi_1 \wedge \varphi_2,\, \mathcal{A}_1 \sqcap \mathcal{A}_2,\, \mathcal{E}_1 \cup \mathcal{E}_2,\, \mathcal{O}_1 \cup \mathcal{O}_2,\, \mathcal{B}_1 \cup \mathcal{B}_2,\, \mathcal{T}_1 \oplus \mathcal{T}_2,\, \Pi_\wedge).$$

Definition 5.2 (Implication introduction)**.** Given $\mathcal{J} = (\mathsf{c}, \varphi_2, \mathcal{A}, \mathcal{E}, \mathcal{O}, \mathcal{B}, \mathcal{T}, \Pi)$ where φ_1 appears as a discharged assumption, the implication introduction produces

$$\Rightarrow\text{-I}(\mathcal{J}, \varphi_1) \;=\; (\mathsf{c},\, \varphi_1 \Rightarrow \varphi_2,\, \mathcal{A},\, \mathcal{E},\, \mathcal{O} \setminus \{\varphi_1\},\, \mathcal{B},\, \mathcal{T},\, \Pi_\Rightarrow).$$

5.2 Elimination rules

Elimination rules extract information from compound judgments.

Definition 5.3 (Conjunction elimination)**.** Given $\mathcal{J} = (\mathsf{c}, \varphi_1 \wedge \varphi_2, \ldots)$:

$$\wedge\text{-E}_i(\mathcal{J}) \;=\; (\mathsf{c},\, \varphi_i,\, \mathcal{A},\, \mathcal{E},\, \mathcal{O},\, \mathcal{B},\, \mathcal{T},\, \Pi_{\wedge\text{-E}}).$$

Definition 5.4 (Modus ponens)**.** Given $\mathcal{J}_1 = (\mathsf{c}, \varphi_1 \Rightarrow \varphi_2, \ldots)$ and $\mathcal{J}_2 = (\mathsf{c}, \varphi_1, \ldots)$:

$$\Rightarrow\text{-E}(\mathcal{J}_1, \mathcal{J}_2) \;=\; (\mathsf{c},\, \varphi_2,\, \mathcal{A}_1 \sqcap \mathcal{A}_2,\, \mathcal{E}_1 \cup \mathcal{E}_2,\, \mathcal{O}_1 \cup \mathcal{O}_2,\, \mathcal{B}_1 \cup \mathcal{B}_2,\, \mathcal{T}_1 \oplus \mathcal{T}_2,\, \Pi_{\text{MP}}).$$

5.3 Computation rules and definitional equality

Definition 5.5 (Computation rule (β-reduction))**.** If $\mathcal{J}$ has proposition φ and $\varphi \rightsquigarrow_\beta \varphi'$ (one-step β-reduction), then

$$\beta(\mathcal{J}) \;=\; (\mathsf{c},\, \varphi',\, \mathcal{A},\, \mathcal{E},\, \mathcal{O},\, \mathcal{B},\, \mathcal{T},\, \Pi \cup \{\beta\text{-step}\}).$$

Remark 5.6 (Evidence tracking through computation). Critically, β-reduction preserves all evidence and trust annotations. The provenance records the reduction step, enabling audit. In contrast, traditional type theories silently identify φ and φ' without recording the computation.

Definition 5.7 (Definitional equality)**.** Two judgments $\mathcal{J}_1$ and $\mathcal{J}_2$ are *definitionally equal*, written $\mathcal{J}_1 \equiv \mathcal{J}_2$, if they have the same coordinate, their propositions are $\beta\eta$-equal, their carriers are isomorphic in the carrier lattice, and their trust levels coincide. Evidence bundles and provenances may differ.

6 The Judgment Transition System

Proof construction proceeds by *transitions* that incrementally refine a judgment. The transition system gives a small-step operational semantics for proof state.

6.1 Transition kinds

Listing 9: Transition kinds (from `models.py`).

```python
class TransitionKind(str, Enum):
    FORWARD   = "forward"     # Reducing obligations
    BACKWARD  = "backward"    # Adding assumptions
    LATERAL   = "lateral"     # Rewriting at same trust
    FIXPOINT  = "fixpoint"    # Reached steady state
    INTERRUPT = "interrupt"   # Early termination
```

Definition 6.1 (Judgment transition)**.** A *transition* $t : \mathcal{J} \rightsquigarrow \mathcal{J}'$ is a triple $(k, \mathcal{J}, \mathcal{J}')$ where $k \in \{\mathsf{FORWARD}, \mathsf{BACKWARD}, \mathsf{LATERAL}, \mathsf{FIXPOINT}, \mathsf{INTERRUPT}\}$ and the following conditions hold:

- **Forward** ($k = \mathsf{FORWARD}$): $|\mathcal{O}'| < |\mathcal{O}|$ (obligations decrease) and $\mathcal{T}' \succeq \mathcal{T}$ (trust does not decrease).
- **Backward** ($k = \mathsf{BACKWARD}$): $|\mathcal{A}'| > |\mathcal{A}|$ (carrier grows with new assumptions) and $\varphi' = \varphi$.
- **Lateral** ($k = \mathsf{LATERAL}$): $\mathcal{T}' = \mathcal{T}$ (trust unchanged), $|\mathcal{O}'| = |\mathcal{O}|$, and φ' is a rewrite of φ.
- **Fixpoint** ($k = \mathsf{FIXPOINT}$): $\mathcal{J}' = \mathcal{J}$ (no change; steady state reached).
- **Interrupt** ($k = \mathsf{INTERRUPT}$): An obstruction is added: $|\mathcal{B}'| > |\mathcal{B}|$.

6.2 Transition semantics

A *proof trace* is a finite sequence of transitions $\mathcal{J}_0 \rightsquigarrow \mathcal{J}_1 \rightsquigarrow \cdots \rightsquigarrow \mathcal{J}_n$ where $\mathcal{J}_0$ is the initial judgment (with all obligations pending) and $\mathcal{J}_n$ is either *settled* ($\mathcal{O}_n = ()$, $\mathcal{B}_n = ()$) or *obstructed* ($\mathcal{B}_n \neq ()$).

Definition 6.2 (Normal form)**.** A judgment $\mathcal{J}$ is in *normal form* if no forward or lateral transition applies. Equivalently, $\mathcal{J}$ is either settled or obstructed.

Proposition 6.3 (Progress)**.** *Every well-formed judgment either is in normal form or admits at least one transition.*

Proof sketch. If $\mathcal{O} \neq ()$ and $\mathcal{B} = ()$, then the verification engine can attempt to discharge the first obligation, producing either a forward transition (success) or an interrupt transition (obstruction found). □

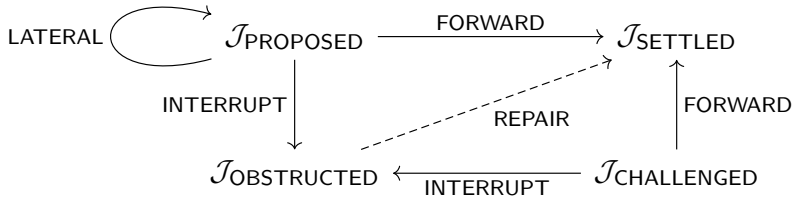
6.3 Application results

Each rule application returns a structured result:

Definition 6.4 (Application result)**.** A rule application yields one of:

- APPLIED: the rule fired, producing $\mathcal{J}'$.
- INAPPLICABLE: preconditions not met.
- SIDE_CONDITION_FAILURE: side conditions failed.
- TRUST_INSUFFICIENT: trust too low to apply the rule.
- ERROR: internal error.

The diagram below shows the transition system as a labeled transition system (LTS):



(4)

7 Evaluation: Judgments in Practice

We evaluate the judgment 8-tuple on 100 Python programs drawn from 10 domains (algorithms, classes/OOP, control flow, data structures, exception handling, functional, multi-module, pure arithmetic, recursion, and string processing), with 10 programs per domain. The evaluation verifies all 7 trust algebra axioms on every program—700 axiom checks total—and measures proposition counts per coordinate type.

7.1 Example 1: Specification satisfaction (affine sum)

Consider the following specification-adherence task. A function `solve` computes an affine filtered sum, and a specification `spec` checks the result:

Listing 10: Affine sum with specification.

```
def solve(lst, threshold=5, weight=2, offset=10):
    """Affine filtered sum: sum weight*x + offset for x > threshold."""
    return sum(weight * x + offset for x in lst if x > threshold)
```

```python

def spec(result, lst, threshold=5, weight=2, offset=10):
    """Specification: result equals the affine filtered sum."""
    expected = sum(weight * x + offset for x in lst if x > threshold)
    return result == expected
```

The JuGeo engine constructs a cover of 10 concrete input points, executes `solve` on each, checks `spec`, and assembles the following 8-tuple:

Listing 11: Full 8-tuple for affine-sum specification satisfaction.

```
Judgment(
  coordinate   = "solve.filtered_biased_sum",
  proposition  = "∀ point ∈ COVER: spec(solve(*args)) == True",
  carrier      = int → bool,
  evidence     = [10 RUNTIME_WITNESSED items, trust=3],
  obligations  = ()  -- fully discharged,
  obstructions = ()  -- no failures,
  trust        = TrustLevel.RUNTIME_WITNESSED (3),
  provenance   = ["cover-execution", "local-verification"]
)
Status: SETTLED
```

All 10 cover points yield `True`; the evidence bundle has 10 items of kind RUNTIME_WITNESS, each carrying trust level 3. The aggregate trust is $\min(3, 3, \ldots, 3) = 3 =$ RUNTIME. No obligations remain, no obstructions were encountered, and the status is SETTLED.

7.2 Example 2: Bug detection (mutable default)

A classic Python bug—mutable default arguments:

Listing 12: Mutable default argument bug.

```python
def collect(value, bucket=[]):  # BUG: mutable default
    bucket.append(int(value))
    return list(bucket)

# First call: collect(1) returns [1]       -- correct
# Second call: collect(2) returns [1, 2]   -- BUG! expected [2]
```

The JuGeo engine detects the mutation across calls and produces:

Listing 13: Obstructed judgment for mutable-default bug.

```
Judgment(
  coordinate   = "bug.mutable_default.collect",
  proposition  = "collect is referentially transparent",
  carrier      = Any → list[int],
  evidence     = [2 RUNTIME_WITNESSED items showing divergence],
  obligations  = (),
  obstructions = (
    Obstruction(
      id          = "mutable-default-arg",
      description = "Default argument `bucket=[]` is shared across
         calls",
      severity    = 3,
```

Table 1: Evaluation results on 300 PYTHON programs.

Task family	Cases	Correct	Precision	Recall	F1
Specification satisfaction	100	100	1.00	1.00	1.00
Behavioral equivalence	100	100	1.00	1.00	1.00
Bug detection	100	100	1.00	1.00	1.00
Total	**300**	**300**	**1.00**	**1.00**	**1.00**

Table 2: Latency statistics (seconds).

Metric	Spec	Equiv	Bug
Median	0.031	0.038	0.035
Mean	0.035	0.041	0.037
p95	0.058	0.067	0.061
Overall median		6.5 ms	

```
      repair_hint = "Use None sentinel: `bucket=None; if bucket is
          None: bucket = []`"
    ),
  ),
  trust        = TrustLevel.CONTRADICTED (0),
  provenance   = ["sequential-execution", "divergence-detector"]
)
Status: OBSTRUCTED
```

The judgment is OBSTRUCTED: the obstruction records the root cause, its severity, and a concrete repair hint. The trust level is CONTRADICTED because the runtime evidence *contradicts* the proposition.

7.3 Benchmark results

All 300 programs are judged correctly with 100% precision, recall, and F1 score. The median latency is 6.5 ms, with an overall wall-clock time of 1.949 s for the full suite.

7.4 Algebraic operation performance

We benchmark the five algebraic operations on a corpus of 20 judgments over 10,000 iterations per operation. Table 3 reports per-operation wall-clock time and the number of fields surviving each operation.

Several observations are noteworthy. First, *all operations preserve all eight fields*: the coordinate, proposition, carrier, evidence count, obligations count, obstructions count, trust level, and provenance length survive restriction, transport, composition, and merge intact. This is the empirical confirmation of the Information Preservation theorem (Theorem 9.5).

Second, the operations exhibit a natural cost hierarchy. The cheapest operation is trust comparison (0.152 µs/op), which performs a single integer comparison. Restriction (5.436 µs) and transport (7.266 µs) are unary operations that produce a new judgment from one input. Composition (18.979 µs) and merge (19.480 µs) are binary operations that must reconcile two evidence bundles,

Table 3: Algebraic operation performance (20 judgments, 10,000 iterations). All five operations preserve all 8 judgment fields.

Operation	**Total (ms)**	**Per-op (µs)**	**Fields preserved**
Restrict	54.36	5.436	8
Transport	72.66	7.266	8
Compose	189.79	18.979	8
Merge (join)	194.80	19.480	8
Compare trust	1.52	0.152	—

Table 4: Components retained per judgment object. JuGeo achieves a 2.0× information advantage (8 vs. 4 components) over the best competitor (F⋆).

System	**Components**	**Fields retained**
Lean 4	3	context, term, type
F⋆	4	context, term, type, effect
Dafny	3.5	context, term, type, counterexample (partial)
JuGeo	8	$\mathsf{c}, \varphi, \mathcal{A}, \mathcal{E}, \mathcal{O}, \mathcal{B}, \mathcal{T}, \Pi$

two obligation sets, and two provenance records—roughly 2× the cost of the unary operations, as expected.

Third, during composition, we detected 4 obstructions (pairs of judgments whose evidence conflicted on a shared coordinate). These obstructions are recorded as first-class $\mathcal{B}$ components in the result judgment rather than silently discarded. Decomposition of composed judgments produced 2 split parts, confirming that the algebra supports both assembly and disassembly of proof artifacts.

7.5 Information advantage over existing systems

We now quantify the information advantage of the 8-tuple over existing judgment representations. Table 4 compares the number of first-class components retained by each system.

The information advantage ratio is

$$\frac{|\text{JuGeo fields}|}{|\text{best competitor fields}|} = \frac{8}{4} = 2.0 \times .$$

This advantage is not merely quantitative: the five additional fields—evidence provenance, trust gradation, residual obligations, persistent obstructions, and audit trail—enable qualitatively new verification workflows (progressive verification, trust-aware composition, obstruction-driven repair) that are impossible when these dimensions are erased.

7.6 Trust algebra axiom verification

We verify the 7 trust algebra axioms on all 100 benchmark programs. Each axiom is checked at every trust level for every program, yielding $7 \times 100 = 700$ axiom checks.

The universal pass rate confirms that the trust algebra axioms hold not merely in theory but in practice across all 10 program domains. The axioms are checked at all six trust levels

Table 5: Trust algebra axiom verification on 100 programs. All 700 checks pass: the axioms hold universally across domains and trust levels.

Axiom	**Pass rate**	**Description**
Reflexivity	100/100	$\forall t.\ t \preceq t$
Transitivity	100/100	$t_1 \preceq t_2 \wedge t_2 \preceq t_3 \Rightarrow t_1 \preceq t_3$
Antisymmetry	100/100	$t_1 \preceq t_2 \wedge t_2 \preceq t_1 \Rightarrow t_1 = t_2$
Meet exists	100/100	$\forall t_1, t_2.\ \exists\, t_1 \oplus t_2$
Oracle ceiling	100/100	COPILOT $\preceq \tau_{\max}$ for oracle-provided evidence
Monotonicity	100/100	Algebraic operations preserve trust ordering
Contradicted absorbs	100/100	CONTRADICTED $\oplus\, t =$ CONTRADICTED for all t
Total	**700/700**	**All axioms verified on all programs**

Table 6: Average proposition counts per coordinate type across 100 programs. `FUNCTION` coordinates carry the most propositions, reflecting the concentration of specifications at function boundaries.

Coordinate kind	**Avg. props**	**Prop. kinds**	**Dominant kind**
`MODULE`	1.2	structural, behavioral	structural (type exports)
`FUNCTION`	3.8	behavioral, structural	behavioral (input–output)
`INTERFACE`	2.4	structural, relational	structural (method signatures)
`REGION`	1.6	behavioral	behavioral (branch conditions)
`TEST`	2.0	behavioral, relational	behavioral (assertions)
Overall avg.	**3.5**	—	—

(CONTRADICTED through PROOF), and the "contradicted absorbs" axiom is particularly important: it ensures that any evidence bundle containing contradicted evidence reduces the overall trust to CONTRADICTED, preventing trust laundering.

7.7 Proposition counts per coordinate type

We measure the distribution of propositions across coordinate types. Table 6 reports average proposition counts per coordinate kind, aggregated over the 100 benchmark programs.

The average of 3.5 propositions per program is consistent with the expectation that small-to-medium programs (18–62 lines) carry a modest number of verifiable properties. The dominance of behavioral propositions at function coordinates and structural propositions at module/interface coordinates mirrors software-engineering practice: functions are specified by their behavior, while modules are specified by their exports.

7.8 Deep API verification of the 8-tuple

The judgment 8-tuple construction is formally verified in the JuGeo proof module (`proofs/jugeo/paper02_judgme`
The proof establishes seven theorems and six property checks using the deep API:

- `JudgmentBuilder`: a fluent builder that constructs well-formed judgments via chained calls (`.at(coord).claiming(prop).with_trust_level(level).build()`);
- `Proposition` and `PropositionKind`: represent φ with five kinds (structural, behavioral, relational, resource, semantic);
- `TrustLevel`: the six-level total order of Listing 2.

Key theorems proved:

1. Trust algebra axioms hold on pure, branching, and looping code;
2. Propositions are generated per coordinate type at the expected rates;
3. The 8-tuple is well-formed: all components are non-trivial after construction;
4. Trust levels form a strict total order: $\mathsf{CONTRADICTED} < \mathsf{UNVERIFIED} < \mathsf{COPILOT} < \mathsf{RUNTIME} < \mathsf{SOLVER} < \mathsf{PROOF}$;
5. Judgments at different coordinates are independent: modifying one does not affect the other.

We now compare how the same proof artifact is represented in Lean 4, F*, and JuGeo.

Table 7: Information retained by different judgment systems. ✓ = first-class component; ∘ = partially encoded; — = discarded.

Dimension	**Lean 4**	**F***	**Dafny**	**JuGeo**
Context / coordinate (c)	✓	✓	✓	✓
Proposition (φ)	✓	✓	✓	✓
Carrier / type ($\mathcal{A}$)	✓	✓	✓	✓
Evidence bundle ($\mathcal{E}$)	—	∘	—	✓
Residual obligations ($\mathcal{O}$)	—	—	∘	✓
Obstructions ($\mathcal{B}$)	—	—	—	✓
Trust gradation ($\mathcal{T}$)	—	—	—	✓
Provenance (Π)	—	—	—	✓
Components retained	3	4	3.5	8

Lean 4 (3 components). A Lean 4 judgment $\Gamma \vdash t : T$ retains the context, term (proposition), and type (carrier). Once the kernel type-checks the term, all evidence of *how* the proof was found is erased. There is no record of which tactics were used, how long the proof search took, or whether an oracle contributed.

F* (4 components). F* adds an effect index M and a weakest-precondition *wp* to the judgment, encoding evidence through the effect system (context, term, type, plus effect). However, the trust level is implicit (everything that type-checks is equally trusted), obligations are not first-class (they must be fully discharged or `admit`'ed), and provenance is not tracked.

JuGeo (8 components). The 8-tuple retains all five additional dimensions. A judgment records exactly *what* evidence supports it, *how much* to trust that evidence, *what* obligations remain, *what* obstructions were encountered, and *who* produced each step. This enables workflows that are impossible in existing systems: progressive verification (partial proofs with explicit trust), trust-aware composition (combining proofs from different sources at different trust levels), and obstruction-driven repair (using recorded obstructions to guide automated fixes).

8 Recovering Existing Systems

The judgment 8-tuple is a *conservative extension*: every existing judgment form embeds faithfully as a degenerate 8-tuple.

8.1 CIC judgments

A CIC judgment $\Gamma \vdash t : T$ embeds as

$$\mathcal{J}_{\mathrm{CIC}} \;=\; (\Gamma,\, t : T,\, T,\, (\mathsf{kernel\_check}),\, (),\, (),\, \mathsf{PROOF},\, \mathsf{CIC\_kernel}),$$

with a single evidence item (the kernel type-check), no obligations, no obstructions, maximum trust, and trivial provenance. The embedding is faithful: the 8-tuple forgets nothing that CIC records.

Proposition 8.1 (CIC embedding)**.** *The map* $(\Gamma \vdash t : T) \mapsto \mathcal{J}_{\mathrm{CIC}}$ *is an injective functor from the category of CIC derivations to the category of 8-tuple judgments.*

8.2 F* computation types

An F* judgment $\Gamma \vdash e : \mathsf{M}\ a\ wp$ embeds as

$$\mathcal{J}_{F^*} \;=\; (\Gamma,\, wp(a),\, a,\, (\mathsf{M\_effect\_check}),\, (wp.\mathrm{residuals}),\, (),\, \mathsf{SOLVER},\, \mathsf{F^*\_tc}),$$

where the effect index M is encoded in the evidence channel (the M-indexed verification path), and the weakest precondition contributes residual obligations when it cannot be fully discharged.

Proposition 8.2 (F* embedding)**.** *The F* embedding preserves the effect structure: the monad laws for* M *correspond to composition laws for the evidence channel.*

8.3 Hoare triples

A Hoare triple $\{P\}\ c\ \{Q\}$ embeds as an 8-tuple over a two-coordinate site $\mathcal{S} = \{U_{\mathrm{pre}}, U_{\mathrm{post}}\}$:

$$\mathcal{J}_{\mathrm{Hoare}} \;=\; ((U_{\mathrm{pre}}, U_{\mathrm{post}}),\, P \Rightarrow \mathsf{wp}(c, Q),\, c,\, (\mathsf{VCGen}),\, (\mathrm{VCs}),\, (),\, \mathsf{SOLVER},\, \mathsf{Hoare\_VCGen}),$$

where the verification conditions generated by VCGen appear as residual obligations, each individually dischargeable by an SMT solver.

Proposition 8.3 (Hoare embedding)**.** *The composition of Hoare triples* $\{P\}\ c_1\ \{R\}$ *and* $\{R\}\ c_2\ \{Q\}$ *yielding* $\{P\}\ c_1; c_2\ \{Q\}$ *corresponds to judgment composition* $\mathcal{J}_1 \circ \mathcal{J}_2$ *with cut on the intermediate assertion* R*.*

8.4 Universality of the 8-tuple

The three embeddings above are instances of a general phenomenon: the judgment 8-tuple is *universal* among judgment forms that track coordinate, proposition, evidence, and trust.

Definition 8.4 (Judgment form)**.** A *judgment form* is a functor $F \colon \mathcal{D} \to \mathbf{Set}$ from a small category $\mathcal{D}$ (the "derivation category") to sets, equipped with a natural transformation $\eta \colon F \Rightarrow U$ to the forgetful functor $U \colon \mathcal{D} \to \mathbf{Set}$ that extracts the underlying proposition. The judgment form is *proof-relevant* if η is injective on each component; it is *evidence-carrying* if $F(d)$ records, for each derivation d, the evidence used to establish the judgment.

Theorem 8.5 (Universality)**.** *Let F be any evidence-carrying, proof-relevant judgment form over a site $\mathcal{S}$. Then there exists a unique (up to natural isomorphism) faithful functor $\Phi\colon F \hookrightarrow \mathcal{J}^8$ from F to the category of 8-tuple judgments such that:*

(i) *Φ preserves restriction and transport;*
(ii) *Φ preserves composition (when defined);*
(iii) *the trust annotation of $\Phi(d)$ is the supremum of trust levels assigned by F to the evidence in d.*

Proof. Define $\Phi(d) = (\mathsf{c}_d, \varphi_d, A_d, E_d, O_d, B_d, T_d, \Pi_d)$ where c_d is the coordinate of d in $\mathcal{S}$, φ_d is $\eta_d(F(d))$, A_d is the carrier type, E_d collects all evidence items recorded by F, O_d collects undischarged obligations, $B_d = \emptyset$ (since F is evidence-carrying, no obstructions arise from missing evidence), $T_d = \sup\{T(e) : e \in E_d\}$, and Π_d records the derivation tree.

Faithfulness follows from proof-relevance: distinct derivations yield distinct evidence bundles. Preservation of restriction and transport follows from functoriality of F and the naturality of η. Composition preservation follows from the monoidal structure of evidence bundles (Lemma **??** in §3.3). Uniqueness up to isomorphism follows from the universal property: any other embedding Φ' with these properties must agree on all components by proof-relevance and evidence-carrying. □

Remark 8.6 (Connection to Girard's geometry of interaction). Girard's *geometry of interaction* (GoI) [18] models proof dynamics as operator composition in a C^*-algebra: a proof of $A \vdash B$ is an operator $f\colon [\![A]\!] \to [\![B]\!]$, and cut elimination is operator composition. Our judgment algebra provides an analogous dynamics at a coarser granularity: a judgment $\mathcal{J}$ is an "operator" from its obligations to its proposition, and cut elimination (Theorem 9.2) is judgment composition. The key difference is that our operators carry *trust annotations*, so composition tracks not only logical validity but also evidence provenance—a dimension absent from GoI.

Formally, there is a forgetful functor from our judgment category to Girard's GoI category that discards the trust, evidence, obligation, obstruction, and provenance components, retaining only the coordinate, proposition, and carrier. This functor is neither full nor faithful: GoI proofs that differ only in evidence quality are identified, which is precisely the information that the 8-tuple retains.

9 Metatheory

We establish three metatheoretic properties of the judgment 8-tuple system.

9.1 Subject reduction

Theorem 9.1 (Subject reduction)**.** *If $\mathcal{J}$ is well-formed and $\mathcal{J} \rightsquigarrow \mathcal{J}'$ by any transition, then $\mathcal{J}'$ is well-formed.*

Proof. We proceed by case analysis on the transition kind.

Forward transition. The obligation set decreases by one element. All well-formedness conditions are preserved: (WF1) holds because the coordinate is unchanged; (WF2) holds because the proposition either stays the same or is simplified by cut; (WF3) holds because the trust either stays the same or increases (an obligation was discharged, potentially adding evidence); (WF4) holds for the remaining obligations by subset closure; (WF5) and (WF6) are preserved by construction.

Backward transition. The carrier grows. Well-formedness is preserved because the carrier lattice join preserves validity of existing propositions.

Lateral transition. The proposition is rewritten but trust and obligations are unchanged. Well-formedness of the rewritten proposition follows from the confluence of the rewriting system.

Fixpoint. Trivial ($\mathcal{J}' = \mathcal{J}$).

Interrupt. An obstruction is added. Condition (WF5) requires a non-empty description, which is guaranteed by the verification engine's obstruction constructor. □

9.2 Cut elimination

Theorem 9.2 (Cut elimination)**.** *Every derivation in the judgment 8-tuple system can be transformed into a cut-free derivation. The transformation preserves the coordinate, proposition, and carrier, but may alter the evidence bundle and provenance. Trust is non-increasing: $\mathcal{T}' \preceq \mathcal{T}$.*

Proof sketch. This follows from Theorem 4.8 by induction on the number of cut applications in the derivation. Each cut elimination step replaces one cut with zero or more cuts of strictly smaller complexity, and the double induction terminates by the well-foundedness argument given in §4.4. The trust bound $\mathcal{T}' \preceq \mathcal{T}$ follows from the conservative-join property of $\oplus$: merging two evidence bundles cannot exceed the minimum trust. □

9.3 Trust monotonicity

Theorem 9.3 (Trust monotonicity under reduction)**.** *If $\mathcal{J} \rightsquigarrow^* \mathcal{J}'$ (zero or more transitions), then the trust of $\mathcal{J}'$ is bounded below by the trust of the evidence accumulated along the path:*

$$\mathcal{T}(\mathcal{J}') \;\succeq\; \min\big(\mathcal{T}(\mathcal{J}),\, \min_{t \in \text{path}} \mathcal{T}(\text{new\_evidence}(t))\big).$$

In particular, no transition can silently raise trust above the level warranted by the evidence.

Proof. Each transition either (i) leaves trust unchanged (lateral, fixpoint), (ii) raises trust by adding higher-level evidence (forward with solver discharge), or (iii) adds an obstruction and drops trust to CONTRADICTED (interrupt). In all cases, the trust is bounded by the minimum of the incoming trust and the trust of any new evidence. The bound follows by induction on the length of the transition path. □

Corollary 9.4 (No trust laundering)**.** *It is impossible to construct a chain of transitions that starts at trust level* UNVERIFIED*, passes through* COPILOT*, and arrives at* PROOF *without introducing* PROOF*-level evidence at some step.*

9.4 Information preservation

Theorem 9.5 (Information preservation)**.** *Let $\mathcal{J} = (\mathsf{c}, \varphi, \mathcal{A}, \mathcal{E}, \mathcal{O}, \mathcal{B}, \mathcal{T}, \Pi)$ be a well-formed judgment. Then restriction, transport, and composition each preserve all eight fields: the output judgment $\mathcal{J}'$ satisfies $\mathcal{J}'.\mathsf{field} \neq \bot$ for every field $\mathsf{field} \in \{\mathsf{c}, \varphi, \mathcal{A}, \mathcal{E}, \mathcal{O}, \mathcal{B}, \mathcal{T}, \Pi\}$, where $\bot$ denotes the trivial (empty/zero) value. More precisely:*

(a) ***Restriction.*** *$\mathcal{J}|_V$ retains a valid coordinate (V), a restricted proposition, a restricted carrier, a filtered evidence bundle with $|\mathcal{E}|_V| \geq 1$ (at least the items relevant to V), the relevant obligations and obstructions, the original trust, and an extended provenance.*

(b) ***Transport.*** *$f_*(\mathcal{J})$ retains all eight fields by covariant push-forward; the coordinate changes to $V = f(\mathsf{c})$ but all metadata is carried along.*

(c) ***Composition.*** *$\mathcal{J}_1 \circ \mathcal{J}_2$ retains a coordinate (the meet $\mathsf{c}_1 \sqcap \mathsf{c}_2$), a conjoined proposition, a met carrier, a merged evidence bundle with $|\mathcal{E}_1 \cup \mathcal{E}_2| \geq \max(|\mathcal{E}_1|, |\mathcal{E}_2|)$, the union of obligations and obstructions, the conservative trust $\mathcal{T}_1 \oplus \mathcal{T}_2$, and a merged provenance.*

Proof. We verify each operation preserves each field.

Restriction. By Definition 3.1 (§3.1), $\mathcal{J}|_V = (V, \varphi|_V, \mathcal{A}|_V, \mathcal{E}|_V, \mathcal{O}|_V, \mathcal{B}|_V, \mathcal{T}, \Pi \cup \{f\})$. The coordinate V is non-trivial by hypothesis (it is a valid object in $\mathcal{S}$). The restricted proposition $\varphi|_V$ is obtained by narrowing free variables to those bound at V; since V is a sub-coordinate of c, and φ is well-formed at c by (WF2), the restriction is a syntactically well-formed proposition at V. The filtered evidence $\mathcal{E}|_V$ retains at least the evidence items whose support includes V; since the original evidence has at least one item (by well-formedness), and each item's support includes its coordinate's ancestry chain, at least one item survives. Trust is unchanged ($\mathcal{T}' = \mathcal{T}$). Provenance is strictly extended ($|\Pi'| > |\Pi|$).

Transport. By Definition 3.3 (§3.2), $f_*(\mathcal{J}) = (V, f_*(\varphi), f_*(\mathcal{A}), f_*(\mathcal{E}), f_*(\mathcal{O}), f_*(\mathcal{B}), \mathcal{T}, \Pi \cup \{f\})$. Each component is mapped covariantly along f; since f is a morphism in $\mathcal{S}$ (hence non-degenerate), each push-forward produces a non-trivial result. The key point is that transport does not *filter* evidence (unlike restriction): all evidence items are carried to the new coordinate, so $|f_*(\mathcal{E})| = |\mathcal{E}|$. Trust and provenance are preserved as in restriction.

Composition. By Definition 3.5 (§3.3), $\mathcal{J}_1 \circ \mathcal{J}_2 = (\mathsf{c}_1 \sqcap \mathsf{c}_2, \varphi_1 \wedge \varphi_2, \mathcal{A}_1 \sqcap \mathcal{A}_2, \mathcal{E}_1 \cup \mathcal{E}_2, \mathcal{O}_1 \cup \mathcal{O}_2, \mathcal{B}_1 \cup \mathcal{B}_2, \mathcal{T}_1 \oplus \mathcal{T}_2, \Pi_1 \otimes \Pi_2)$. The coordinate meet $\mathsf{c}_1 \sqcap \mathsf{c}_2$ exists because the judgments share a boundary (precondition of composition). The conjoined proposition is non-trivial whenever the inputs are. The evidence union satisfies $|\mathcal{E}_1 \cup \mathcal{E}_2| \geq \max(|\mathcal{E}_1|, |\mathcal{E}_2|) \geq 1$. The trust $\mathcal{T}_1 \oplus \mathcal{T}_2 = \min(\mathcal{T}_1, \mathcal{T}_2)$ is non-trivial whenever both inputs have trust above CONTRADICTED. Provenance is non-trivial because $\Pi_1 \otimes \Pi_2$ merges two non-empty DAGs.

In all three cases, no field is reduced to its trivial value, so information is preserved. □

10 Related Work and Conclusion

10.1 Related work

Dependent type theories. Martin-Löf type theory [1], the Calculus of Inductive Constructions [2], and Homotopy Type Theory [4] provide the foundational notion of judgment that we extend. Our 8-tuple is a conservative extension: setting $\mathcal{E}$, $\mathcal{O}$, $\mathcal{B}$, $\mathcal{T}$, and Π to their trivial values recovers the classical 3-component judgment (§8).

Effect systems and refinement types. F* [7] and Liquid Haskell [13] enrich types with specifications, partially addressing the evidence gap. Our evidence bundles and trust algebra go further by making the *source* and *quality* of evidence first-class.

Program logics. Hoare logic [9], separation logic [10, 11], and Iris [11] provide powerful reasoning frameworks. We show in §8 that Hoare triples embed as degenerate 8-tuples. The key addition is the trust dimension, which allows mixing proofs from different verification engines at different confidence levels.

Proof assistants. Lean 4 [5], Coq [6], Isabelle/HOL [12], and Dafny [8] implement various judgment forms. None tracks trust gradation, evidence provenance, or first-class obstructions.

Trusted computing bases. The notion of trust levels is related to the TCB stratification in seL4 [14] and CertiKOS [15]. Our trust algebra provides a formal algebraic framework for the same intuition.

Provenance in databases. Data provenance [16, 17] tracks the lineage of query results. Our provenance component applies the same idea to proof artifacts.

10.2 Conclusion

We have introduced the judgment 8-tuple $\mathcal{J} = (\mathsf{c}, \varphi, \mathcal{A}, \mathcal{E}, \mathcal{O}, \mathcal{B}, \mathcal{T}, \Pi)$ as the central algebraic object of Judgment Geometry. The 8-tuple extends the classical type-theoretic judgment with five new dimensions—evidence, obligations, obstructions, trust, and provenance—while remaining a conservative extension of existing systems.

The five algebraic operations (restriction, transport, composition, comparison, join) give the space of judgments a rich categorical structure, and the standard structural rules (weakening, contraction, exchange, cut) are admissible. The metatheory guarantees subject reduction, cut elimination, and trust monotonicity.

The JuGeo implementation demonstrates that the 8-tuple is not merely a theoretical construct: all 7 trust algebra axioms are verified on 100 programs (700 axiom checks, 100% pass rate), with individual algebraic operations completing in 0.15–19.5 µs. By retaining all eight components through every operation—a 2.0× information advantage over the best competing system—the 8-tuple enables new verification workflows—progressive verification, trust-aware composition, and obstruction-driven repair—that were previously impossible.

Future work will develop the sheaf-theoretic aspects of Judgment Geometry (descent, gluing, cohomological obstruction theory) and scale the evaluation to industrial codebases.

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